

Wearable Robotics

Brushed DC Motors

Prof. Edgar Bolívar

Sep. 20th, 2023

Brushed motors:

Pros:

Simple to implement and model
(PWM signal + H-bridge)

Cons: (compared to brushed)

- Greater friction from brushes
- Lower speeds
- Lower specific power (W/kg)

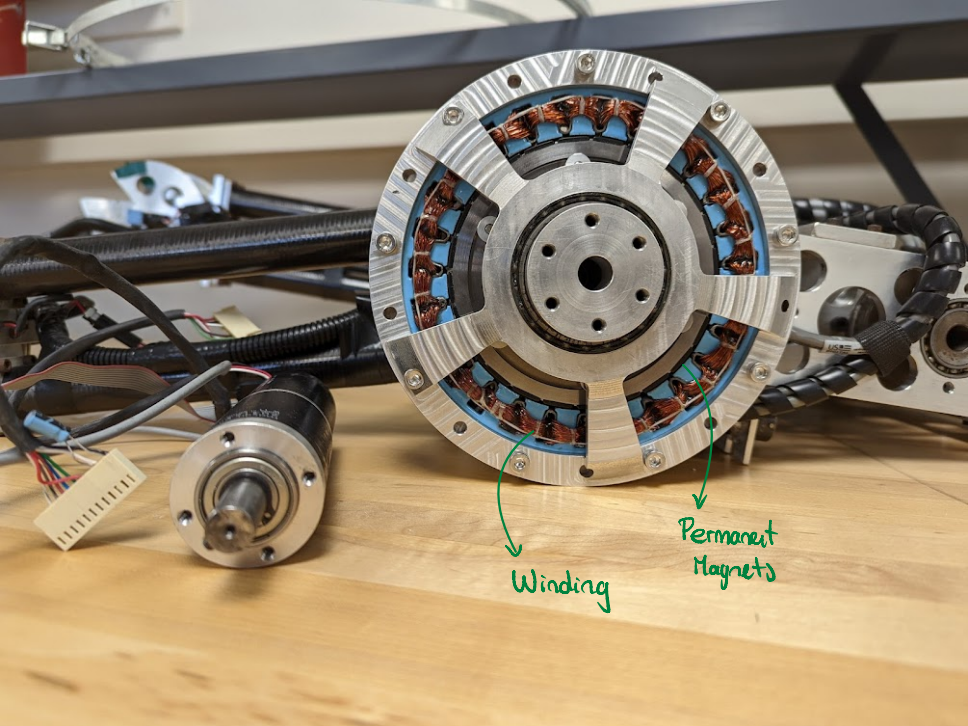
Brushless motors:

Pros:

- Less friction
- High speed and specific power

Cons:

- Requires electrical commutation
(High computing power is required)



Winding

Permanent
Magnets

DC Brushed Motors:

Rules of thumb

Back-EMF voltage is proportional to speed

$$V_{\text{emf}} = k_v \dot{\theta}_m$$

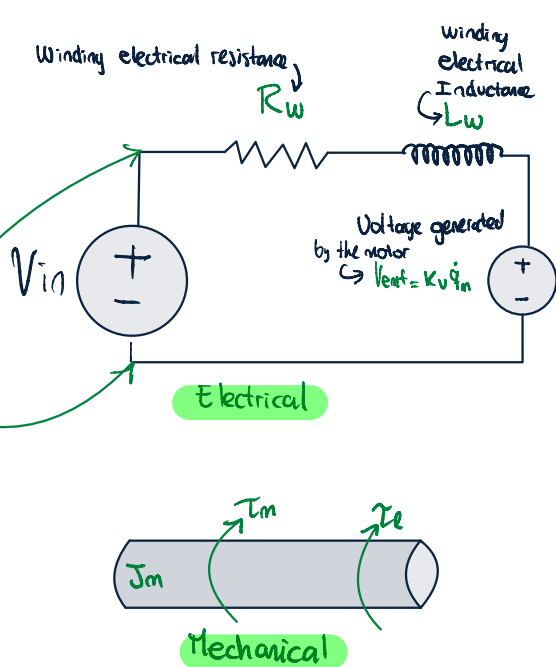
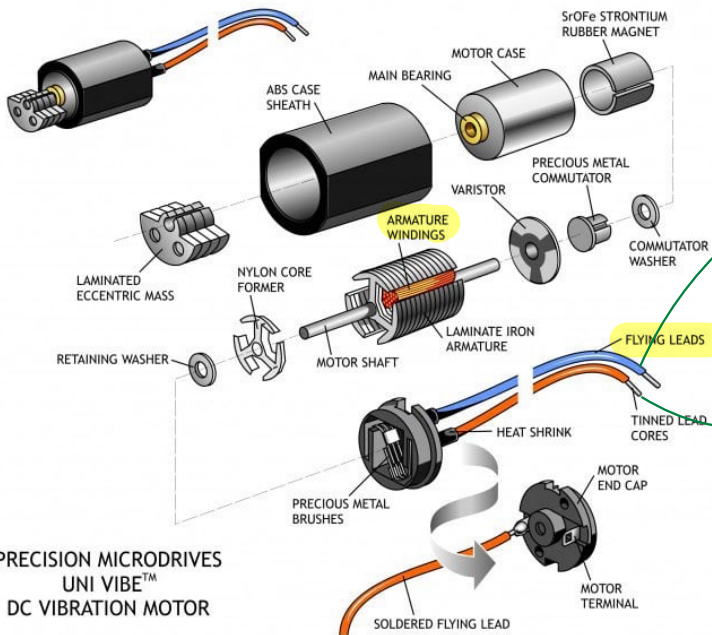
Voltage constant $\left[\frac{\text{V} \cdot \text{s}}{\text{rad}} \right]$

Current is proportional to torque

$$\tau_m = k_T i_m$$

Torque constant $[\text{Nm/A}]$

Mechatronic model of a permanent-magnet brushed motor



KVL:

$$-V_{in} + V_{Rw} + V_{Lw} + V_{emf} = 0$$

$$V_{in} = i R_w + L_w \frac{di}{dt} + K_v \dot{\theta}_m$$

Newton-Euler:

$$J_m \ddot{\theta}_m = \sum M$$

Electromagnetic torque

$$= \tau_m + \tau_l - b_m \dot{\theta}_m$$

$$J_m \ddot{\theta}_m = \tau_m + \tau_l - b_m \dot{\theta}_m$$

$$= K_t i - b_m \dot{\theta}_m + \tau_l$$

Motor constants:

Back-EMF / Voltage constant

$$V_{emf} = K_v \dot{\theta}_m$$

$$\tau_m = K_t i$$

$K_v = K_t$ (in SI units)

Torque constant

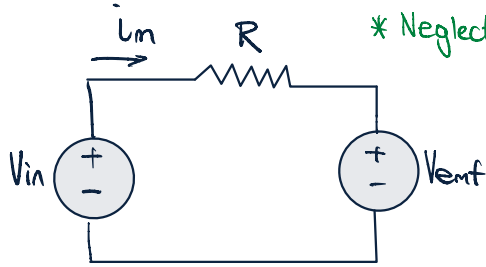
Many torque sources can be modeled, e.g., Coulomb friction

Load torque

Viscous friction

The torque-velocity relationship

Electric motors behave as actuators and generators simultaneously



* Neglect the inductor for simplicity

Recall that

$$\tau_m = k_t i_m$$

What happens to the current when the motor rotates?

$$V_{emf} = k_v \dot{\theta}_m \rightarrow \text{generator.}$$

The higher the motor speed, the less current will go through the winding (less torque)

From KVL:

$$-V_{in} + i_m R + k_v \dot{\theta}_m = 0$$

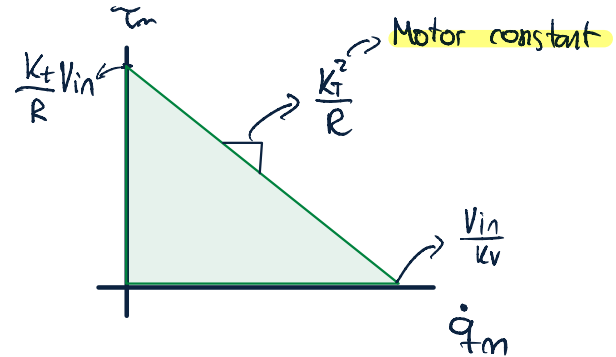
$$\frac{\tau_m R}{k_t} = V_{in} - k_v \dot{\theta}_m$$

* $k_v = k_t$
SI units

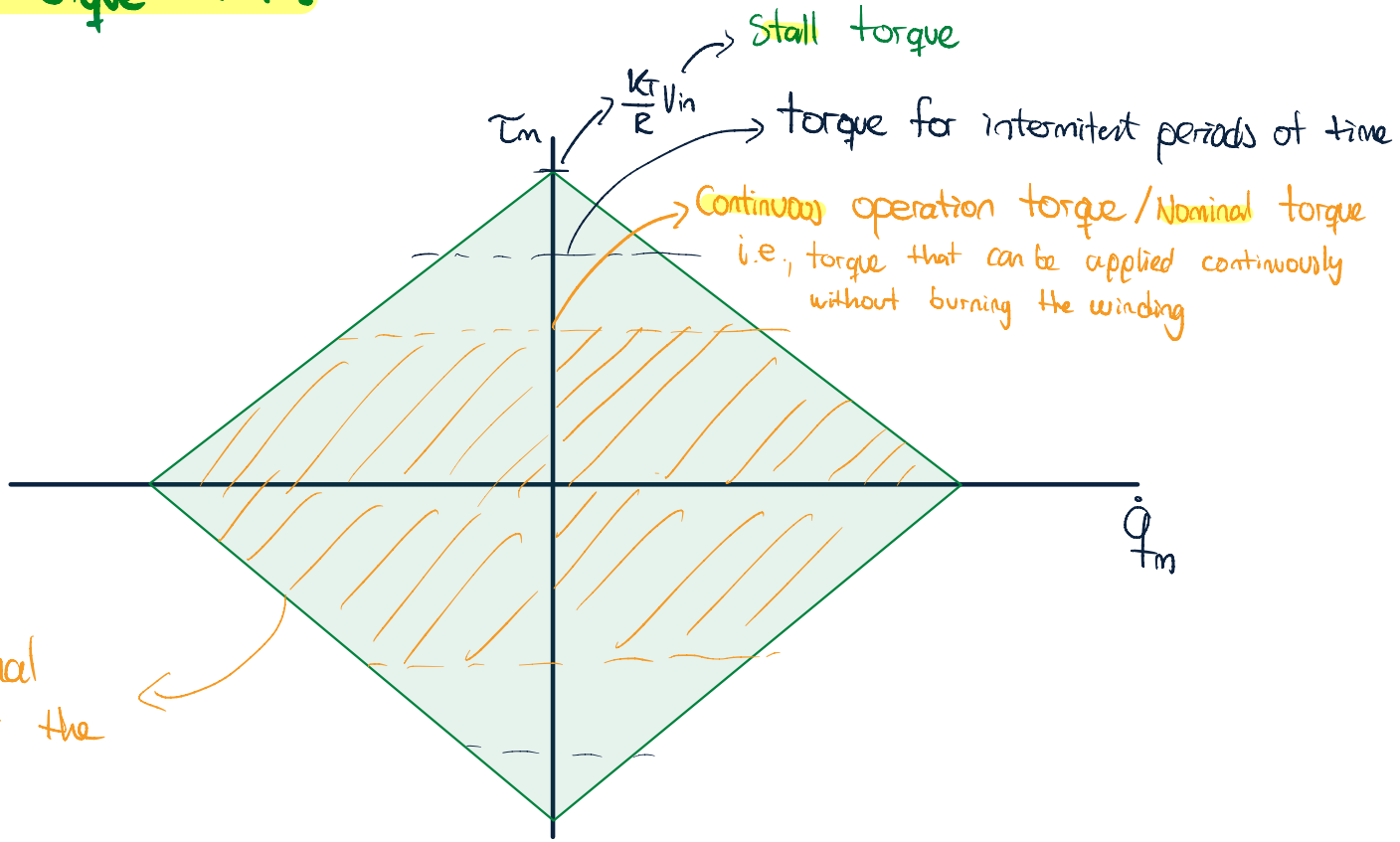
Torque relates

linearly with speed.

$$\tau_m = \frac{k_t}{R} V_{in} - \frac{k_v k_t}{R} \dot{\theta}_m$$



Different torque limits



Example:

What is the input voltage and winding current for a DC-motor ($R_w = 0.2 \Omega$, $k_T = 0.1 \frac{\text{Nm}}{\text{A}}$, $b_m = 0.02 \frac{\text{N}\cdot\text{m}\cdot\text{s}}{\text{rad}}$) rotating at a constant speed of 20 rad/sec with a constant load torque of -2 Nm ?

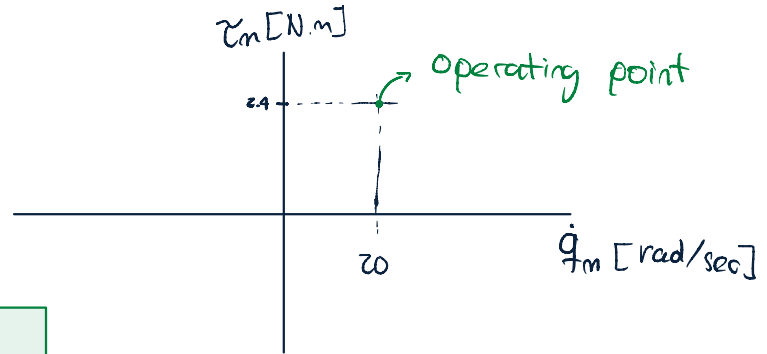
1) From the mechanical domain

$$\underbrace{J \ddot{\varphi}_m}_{0 \text{ (constant speed)}} = \underbrace{\tau_m}_{k_T i_m} - b_m \dot{\varphi}_m + \tau_L$$

$$i_m = \frac{-(0.02)(20) - 2}{-k_T}$$

$$i_m = 24 \text{ A}$$

$$\tau_m = k_T i_m = 2.4 \text{ Nm}$$



2) From the electrical domain

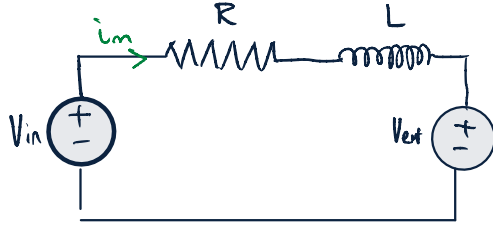
$$V_{in} = R_w i_m + \underbrace{L \frac{di}{dt}}_{0 \text{ (steady state)}} + V_{emf}$$

$$= (0.2)(24) + (0.1)(20)$$

$$V_{in} = 6.8 \text{ V}$$

Kahoot!

Energy consumption of electric motors



Power delivered by the power supply: $V_{in} i_m$ [W]

Energy consumed between t_0 and t_f :

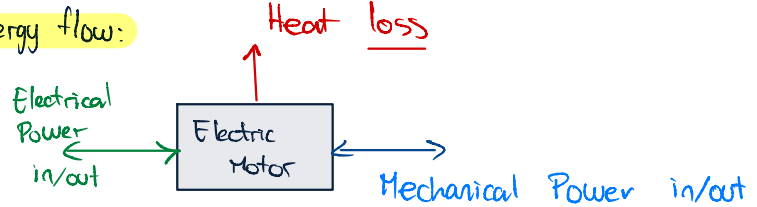
$$E = \int_{t_0}^{t_f} V_{in} i_m dt$$

$$= \int_{t_0}^{t_f} (i_m R + L \frac{di_m}{dt} + V_{ext}) i_m dt$$

Electrical energy Heat loss Mechanical output energy
(can include mechanical losses)

$$E_m = \int_{t_0}^{t_f} (i_m^2 R + \dot{q}_m \tau_m) dt$$

Energy flow:



Typical cases:

Actuator: Electrical ↗ Mechanical
 ↘ Heat loss

Generator*: Mechanical ↗ Electrical
 ↘ Heat loss

* It is possible for mechanical power to flow into the motor and still require electrical power to flow into the motor *

PWM and the role of the inductor

The strategy to control input motor voltage (V_{in}) is through Pulse Width Modulation (PWM)

Why? Transistors are excellent to switch on/off.
Changing the input voltage at high frags ($< 1\text{kHz}$) is challenging without transistors.

What is PWM?

A signal with on/off states defined by

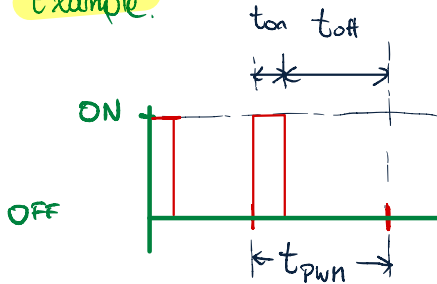
Duty Cycle

$$\frac{t_{on}}{t_{pwm}} \cdot 100$$

Frequency

$$F_{PWM} = \frac{1}{t_{pwm}}$$

Example.

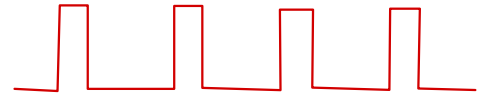


* If power supply voltage is 10V
a PWM duty cycle is equivalent to
a 1V.

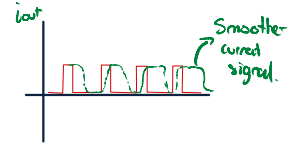
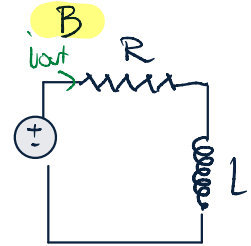
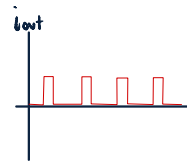
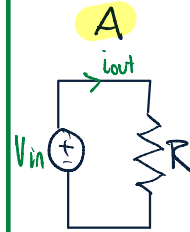
Role of the inductor

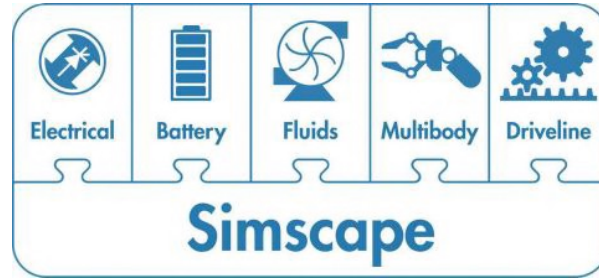
(Smooth input current)

Input voltage (V_{in})



Output current





Multi-domain models — Resources

- Simulink on ramp
- Simscape on ramp

Modeling winding temperature

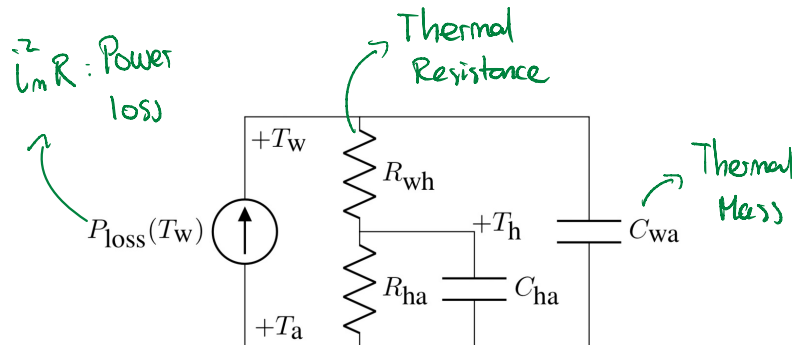


Figure 2: Thermal model of the motor's winding. It describes winding (T_w) and housing (T_h) temperatures as a function of the Joule heating generated by the motor current. In this electrical analogy, temperature is equivalent to voltage and heat flux is equivalent to current, as discussed in [20, 22, 23].

Winding Resistance changes with temperature

$$R_m = R_{m@a}(1 + \alpha_{Cu}(T_w - T_a)),$$

Winding Resistance $\downarrow$ Winding Resistance at known temperature $\downarrow$ Copper's temperature coefficient

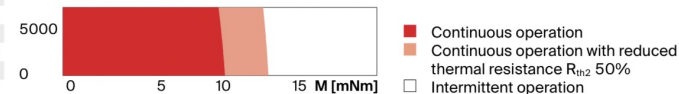
$\alpha = 0.00393 \left[\frac{1}{K} \right]$

HEBI IT!

Maxon Motor DCX 19s

Motor Data								
1_	Nominal voltage	V	4.5	6	9	12	18	24
2_	No load speed	rpm	6440	6350	6260	6360	6360	6350
3_	No load current	mA	72	53	34.6	26.5	17.7	13.2
4_	Nominal speed	rpm	5080	4540	4350	4490	4490	4480
5_	Nominal torque	mNm	7.46	10.3	10.8	11.0	11.0	10.9
6_	Nominal current (max. continuous current)	A	1.20	1.20	0.829	0.643	0.428	0.319
7_	Stall torque	mNm	35.7	36.3	35.8	38.0	37.8	37.5
8_	Stall current	A	5.42	4.07	2.64	2.13	1.41	1.05
9_	Max. efficiency	%	78	79	79	79	79	79
10_	Terminal resistance	Ω	0.831	1.47	3.40	5.63	12.7	22.8
11_	Terminal inductance	mH	0.045	0.082	0.191	0.329	0.740	1.320
12_	Torque constant	mNm/A	6.58	8.90	13.5	17.8	26.7	35.6
13_	Speed constant	rpm/V	1450	1070	705	536	358	268
14_	Speed/torque gradient	rpm/mNm	183	177	177	170	170	172
15_	Mechanical time constant	ms	5.12	4.99	4.92	4.89	4.89	4.90
16_	Rotor inertia	gcm²	2.67	2.68	2.65	2.75	2.74	2.72
Thermal data			Operating Range					
17_	Thermal resistance housing-ambient	K/W	176	n [rpm]		Winding 9 V		
18_	Thermal resistance winding-housing	K/W	6.5					
19_	Thermal time constant winding	s	11.6					
20_	Thermal time constant motor	s	312					
21_	Ambient temperature ball bearings	°C	-40...+85	20000				
	Ambient temperature sleeve bearings	°C	-30...+85					
22_	Max. winding temperature	°C	100	15000				
Mechanical data ball bearings								
23_	Max. speed	rpm	7500	10000				
24_	Axial play	mm	0...0.1					
	Preload	N	2.5					
25_	Radial play	mm	0.02	5000				
26_	Max. axial load (dynamic)	N	2.5					
27_	Max. force for press fits (static)	N	30					
	(static, shaft supported)	N	440					
28_	Max. radial load [mm from flange]	N	16 [5]					
Mechanical data sleeve bearings			Modular System			Details on catalog page 36		
23_	Max. speed	rpm	7500	Gear	Stages [opt.]	Sensor	Motor Control	
24_	Axial play	mm	0...0.2	371_GPX 19 A/C	1-2 [3-4]	473_ENX 10 EASY	532_ESCON Module 24/2	
	Preload	N	0	372_GPX 19 LN/LZ	1-2 [3-4]	473_ENX 10 QUAD	532_ESCON 36/2 DC	
25_	Radial play	mm	0.02	373_GPX 19 HP	2-3 [4]	474_ENX 10 EASY XT	541_EPOS4 Micro 24/5	
26_	Max. axial load (dynamic)	N	0.1	375_GPX 22 A/C	3-4	475_ENX 16 EASY	542_EPOS4 Module 24/1.5	
27_	Max. force for press fits (static)	N	80	376_GPX 22 LN/LZ	3-4	476_ENX 16 EASY XT	543_EPOS4 Compact 24/5 3-axe	
	(static, shaft supported)	N	440	377_GPX 22 HP	4	477_ENX 16 EASY Abs.	544_EPOS4 Compact 24/1.5	
28_	Max. radial load [mm from flange]	N	3 [5]			478_ENX 16 EASY Abs. XT		
						486_ENX 16 RIO		
Other specifications								
29_	Number of pole pairs		1					
30_	Number of commutator segments		9					
31_	Weight of motor	g	50					
32_	Typical noise level	dBA	48	Configuration				

Bearing: Ball bearings preloaded/sleeve bearings
Commutation: Precious metal brushes with Cu / graphite brushes



Scrape of many motor parameters (multy manufacturer)

<https://github.com/ekrimsk/MGDB> -

Erez Krinsky

Selection of DC-motor (Assignment)

Goal: Select the motor that will guarantee operation while minimizing

- mass
- cost
- Energy

Steps:

1) Get the required load $\rightarrow$ may require a dynamic process on its own

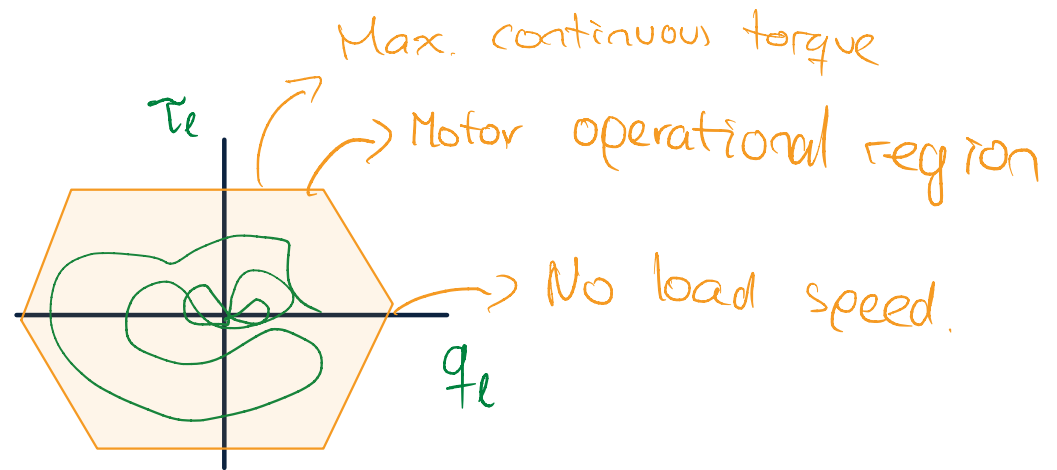
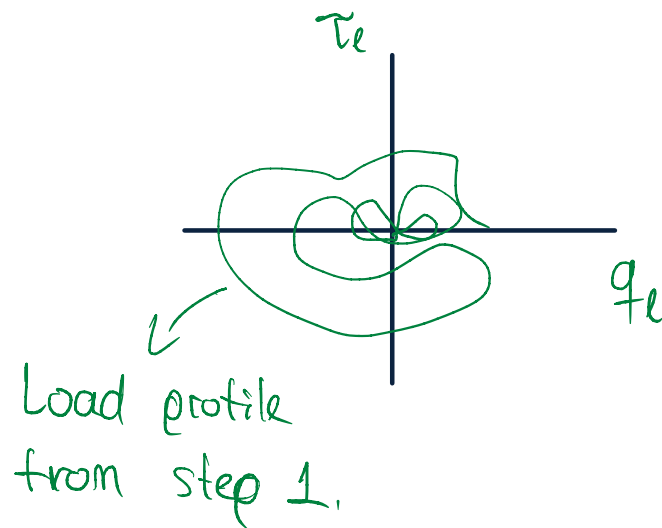
2) Have an idea of the motor size by calculating the maximum load power

3) Make sure that the motor speed and torque can accomplish the task.

1a) use the mechanical motor equation to solve for winding current from load torque and motor kinematics

1b) Use the electrical-domain motor eq. to solve for input voltage (PWM)

Caution! Your motor may not achieve the task even if motor power is higher than load power



4) Iterate until you find the motor that satisfies your goal